Network Traffic Incident Analysis and PCAP Triage Report

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Environment	Controlled lab environment
Primary tool	Wireshark
Capture file	Lab 4 Attack Analysis.pcapng
Assessment focus	PCAP triage, protocol analysis, endpoint review, and incident documentation

Executive Summary

This report presents the results of a network traffic analysis conducted in Wireshark using a captured PCAP file associated with an incident response scenario. The purpose of the assessment was to review packet-level evidence, identify noteworthy protocol activity, and document observations that may support triage and follow-up investigation.

The capture contained 83,782 packets over approximately 921.823 seconds. Traffic was dominated by communications involving hosts 192.168.30.128 and 192.168.30.130, with additional activity involving 192.168.30.1, 192.168.30.2, 192.168.30.254, multicast destinations, and local broadcast traffic. Observed protocols included HTTP, TLS, SMTP, SMB-related traffic, ICMP, ARP, IPv4, and IPv6 multicast activity.

The packet review identified noteworthy web requests, encrypted-session negotiation, SMTP communication attempts that did not complete successfully, SMB-related exchanges, and local ARP requests. The evidence supports further investigation, but the packet data is most accurately described as suspicious or relevant network activity rather than definitive proof of a specific attack technique without corroborating endpoint or log evidence.

Introduction

This assessment was conducted to analyze a provided PCAP file associated with a reported security incident. The objective was to inspect captured traffic in Wireshark, extract relevant host and protocol information, and identify behaviors that may assist with incident triage, scoping, and follow-up investigation.

Wireshark was used as the primary analysis platform because it supports packet inspection, protocol filtering, endpoint review, and examination of individual conversations. The workflow focused on practical traffic analysis and evidence handling rather than theoretical discussion of the tool.

Scope

This review was limited to the provided packet capture file, identified in the original lab as “Lab 4 Attack Analysis.pcapng.” The analysis covered file metadata, capture properties, local addressing information,

HTTP requests, TLS handshakes, SMTP traffic, TCP conversation evidence, endpoint distribution, SMB-related activity, and ARP responses. The assessment was conducted in a controlled environment for technical documentation and incident triage review.

Methodology

The PCAP file was loaded into Wireshark and reviewed through a staged analysis process. Initial review focused on file metadata, capture duration, system details, and packet statistics. Subsequent analysis used display filters to isolate HTTP requests, TLS handshake traffic, SMTP communication, TCP session behavior, endpoint summaries, SMB-related traffic, and ARP responses.

Observations were then documented conservatively, distinguishing between packet-level evidence directly visible in the capture and stronger conclusions that would require confirmation from endpoint logs, firewall records, email server logs, or other corroborating artifacts.

Findings and Analysis

1. Capture File Properties

The source PCAP file is approximately 13 MB and contains 83,782 packets over a timespan of 921.823 seconds, with Ethernet used as the encapsulation method. The first packet was captured on 2020-08-02 23:21:29 and the last on 2020-08-02 23:36:51. The capture system is identified as Linux 5.5.0-kali2-amd64 using Dumpcap (Wireshark) 3.2.3 on AMD Ryzen 5 3600 hardware.

2. Local Host and Gateway Information

The local virtual machine address observed in the source material is 192.168.30.128 with a gateway of 192.168.30.1. These values establish the local segment and help frame communications between the most active hosts in the capture.

3. HTTP Request Activity

HTTP request filtering showed noteworthy web activity in the capture, including retrieval of robots.txt. That observation is relevant because it may reflect reconnaissance or application discovery behavior, although it should be documented as noteworthy traffic rather than presumed malicious activity by itself.

4. TLS Handshake Activity

TLS handshake filtering identified encrypted-session negotiation on the network. The presence of TLS client hello traffic confirms attempts to establish encrypted communications between endpoints and may support correlation with later encrypted sessions.

5. TCP Conversation Evidence

TCP filtering showed session-establishment behavior consistent with standard client-server negotiation, including SYN traffic between internal hosts and multiple destination ports. This is useful for identifying active connection attempts and prioritizing endpoints for further review.

6. SMTP Communication Attempts

SMTP filtering showed repeated communication attempts between 192.168.30.128 and 192.168.30.130 involving commands such as EHLO and AUTH NTLM, with several server responses indicating command or syntax errors and authentication not enabled. The evidence supports documentation of unsuccessful SMTP activity that warrants follow-up review.

7. IPv4 and IPv6 Endpoint Distribution

Endpoint summaries show that most activity involved 192.168.30.128 and 192.168.30.130, with smaller volumes involving 192.168.30.2, 192.168.30.1, 192.168.30.254, and multicast destinations such as 239.255.255.250 and 224.0.0.x addresses. Limited IPv6 multicast activity was also present and is consistent with local service-discovery behavior.

8. SMB and ARP Activity

The packet set includes SMB-related traffic and ARP requests on the local segment. These are relevant triage observations because they may support later analysis of file-sharing behavior, host interaction, or address-resolution anomalies. However, the presence of these packets alone does not confirm malicious activity without additional evidence.

Key Observations

- The capture includes 83,782 packets over approximately 15 minutes of observed traffic.
- The most active systems in the capture are 192.168.30.128 and 192.168.30.130.
- HTTP, TLS, SMTP, SMB-related, ICMP, ARP, IPv4, and IPv6 multicast traffic are all present.
- Notable traffic includes retrieval of robots.txt, TLS client hello exchanges, unsuccessful SMTP interactions, and internal connection attempts to multiple ports.
- The packet evidence supports further investigation, but stronger conclusions about specific attack types would require corroborating endpoint or infrastructure logs.

Recommendations

- Prioritize review of communications between 192.168.30.128 and 192.168.30.130, since they account for the majority of traffic in the capture.
- Correlate the PCAP findings with endpoint logs, firewall logs, proxy logs, and email server logs before concluding that specific attack types occurred.
- Review HTTP request targets, including robots.txt, to determine whether the activity reflects normal browsing, reconnaissance, or scripted access.
- Investigate unsuccessful SMTP traffic to determine whether it reflects misconfiguration, blocked communications, or attempted outbound transfer.
- Review SMB-related traffic and ARP behavior for anomalies such as unexpected responder behavior, inconsistent MAC mappings, or suspicious file-sharing patterns.
- Preserve the PCAP hash values and file metadata as part of the incident evidence record.

Conclusion

This assessment reviewed a captured PCAP file in Wireshark to support triage of a reported security incident. The analysis identified a variety of relevant network behaviors, including HTTP requests, TLS negotiation, SMTP communication attempts, SMB-related traffic, local ARP responses, and concentrated traffic between a small number of internal hosts.

The packet evidence supports deeper investigation, especially involving hosts 192.168.30.128 and 192.168.30.130, but the most defensible conclusion is that the PCAP contains suspicious and noteworthy traffic rather than definitive proof of a specific attack technique without corroborating evidence. This framing makes the report more accurate, more defensible, and more suitable for professional incident analysis.

Assessment Snapshot

Analysis Area	Primary Evidence	Assessment Value
Capture overview	File properties and metadata	Establishes timeline, integrity context, and capture environment
Host focus	192.168.30.128 and 192.168.30.130	Identifies priority endpoints for further investigation
Web activity	HTTP requests and robots.txt retrieval	Supports review of web-related behavior and potential reconnaissance
Encrypted traffic	TLS handshake activity	Confirms attempts to establish encrypted sessions
Email traffic	SMTP session attempts with errors	Highlights unsuccessful mail-related activity requiring follow-up
Internal communications	TCP session attempts and SMB-related traffic	Supports scoping of host interaction and service access patterns
Local segment activity	ARP requests and IPv4/IPv6 endpoint summaries	Describes local network context and potential anomaly areas

Appendix A: Assessment Evidence

The following figures provide supporting technical evidence for the activities and observations documented in this report.

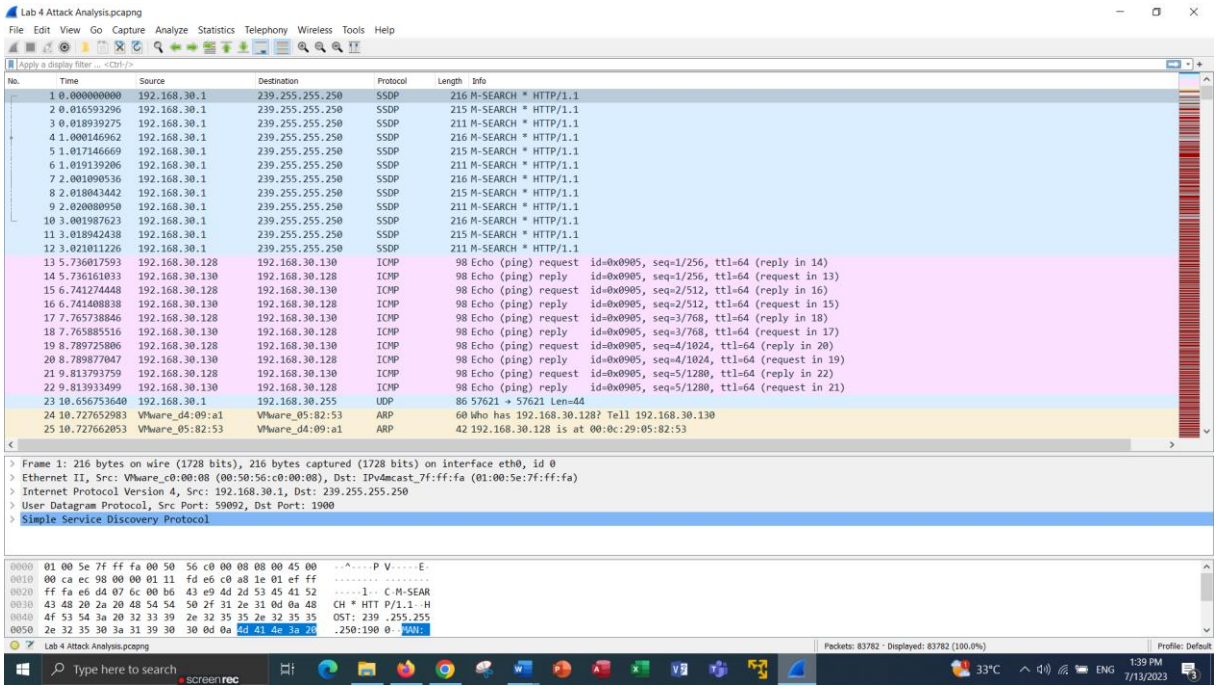

Figure A1. Wireshark view of the loaded PCAP file showing overall packet activity, ICMP traffic between 192.168.30.128 and 192.168.30.130, and ARP requests on the local segment.

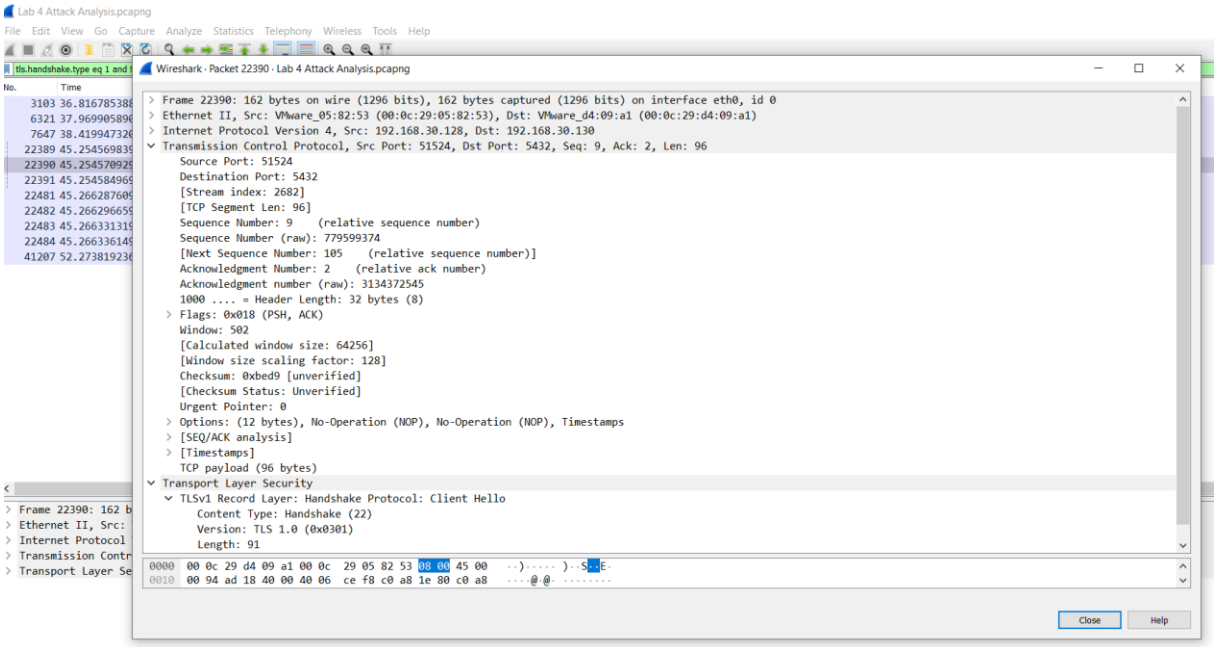

Figure A2. File properties dialog for the PCAP, showing capture metadata including file size, hash values, capture duration, operating system, hardware, and packet statistics.

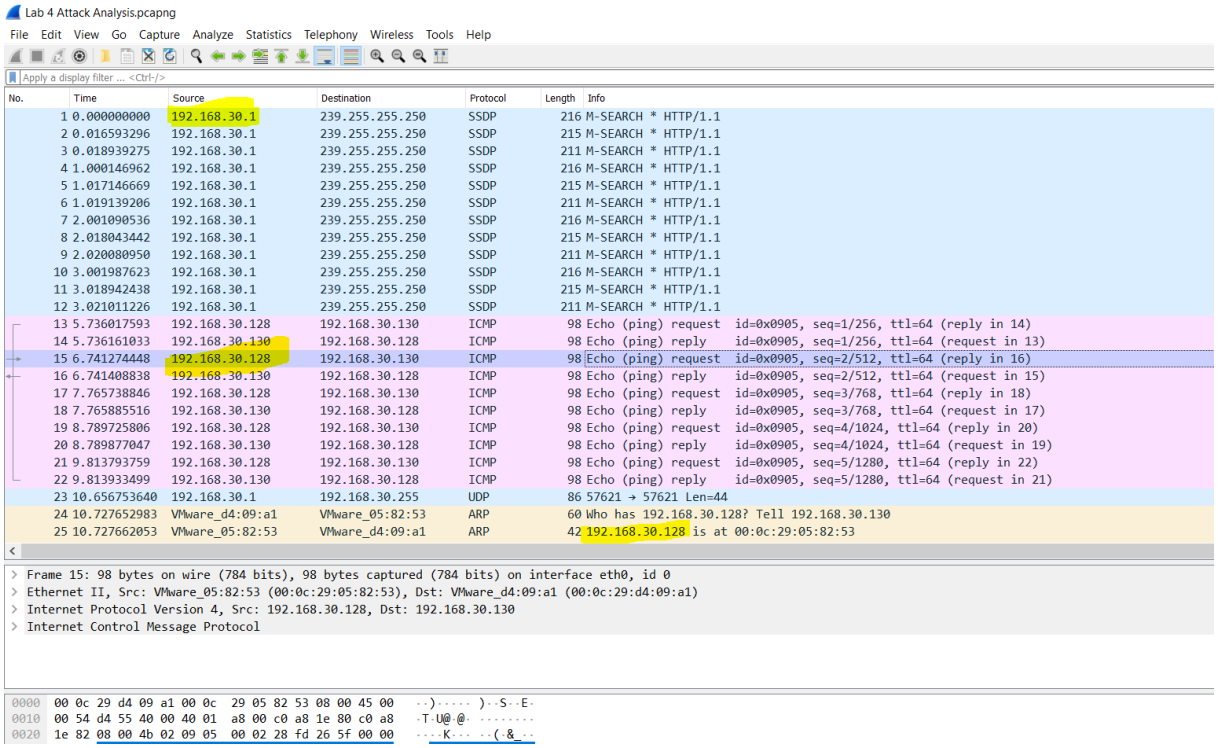

Figure A3. Wireshark evidence highlighting local addressing information, including the gateway 192.168.30.1 and the local host 192.168.30.128, alongside ICMP and ARP traffic.

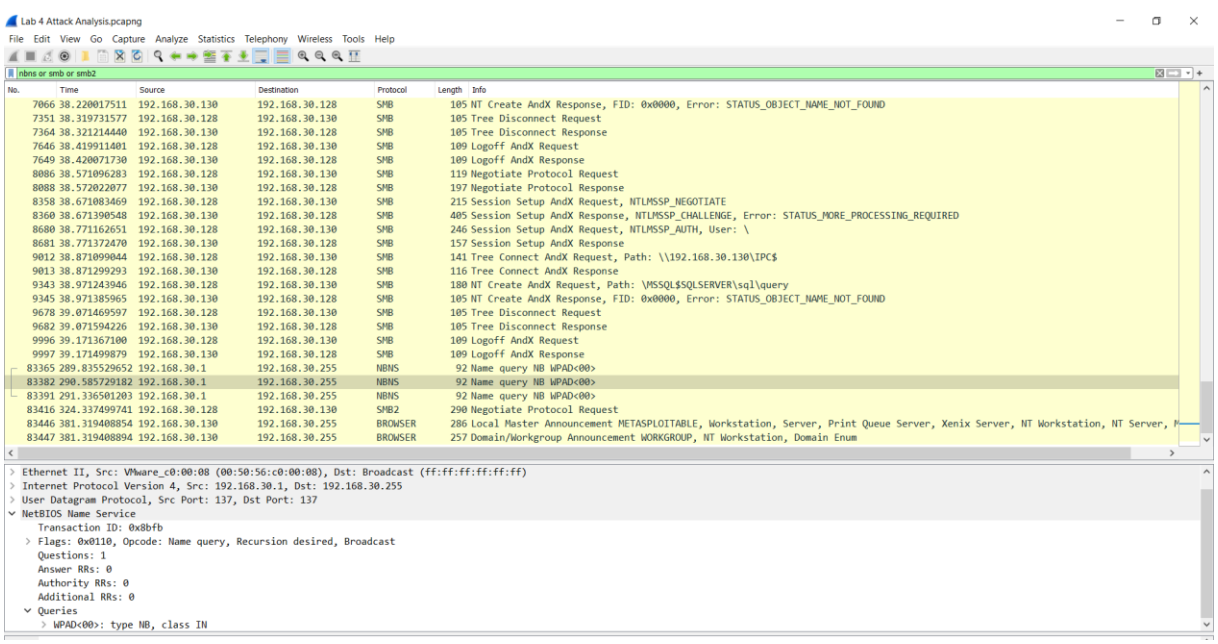

Figure A4. SMB-related traffic filtered in Wireshark, showing AndX request activity relevant to file-sharing or Windows network communication review.

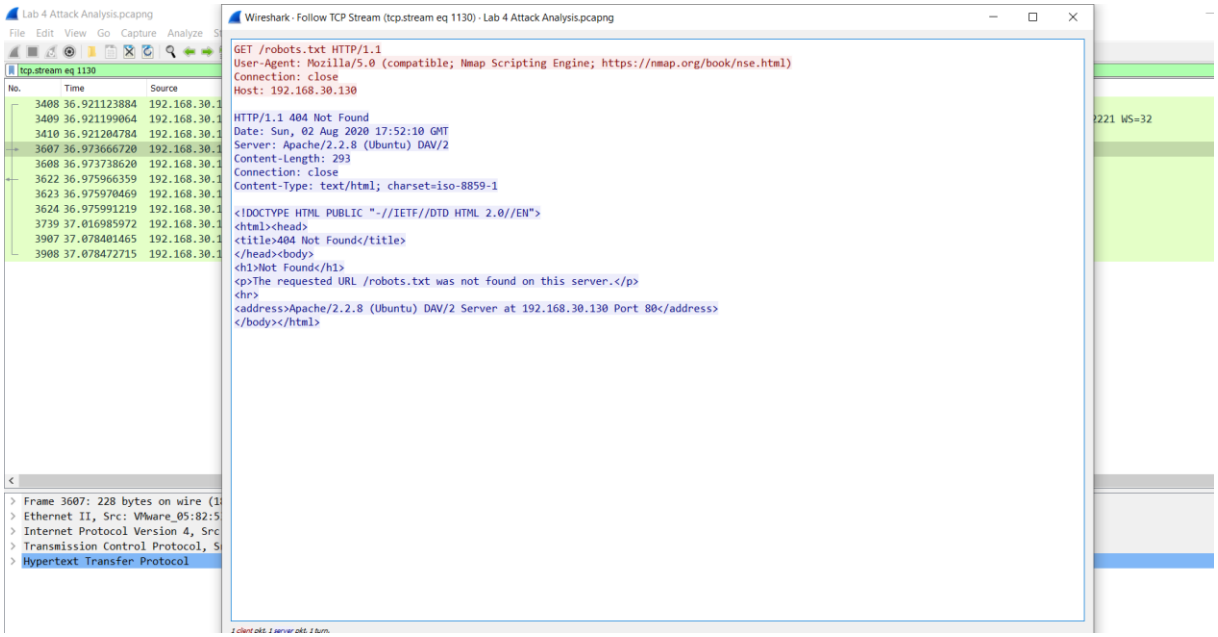

Figure A5. HTTP request filtering results showing web activity and retrieval of robots.txt within the capture.

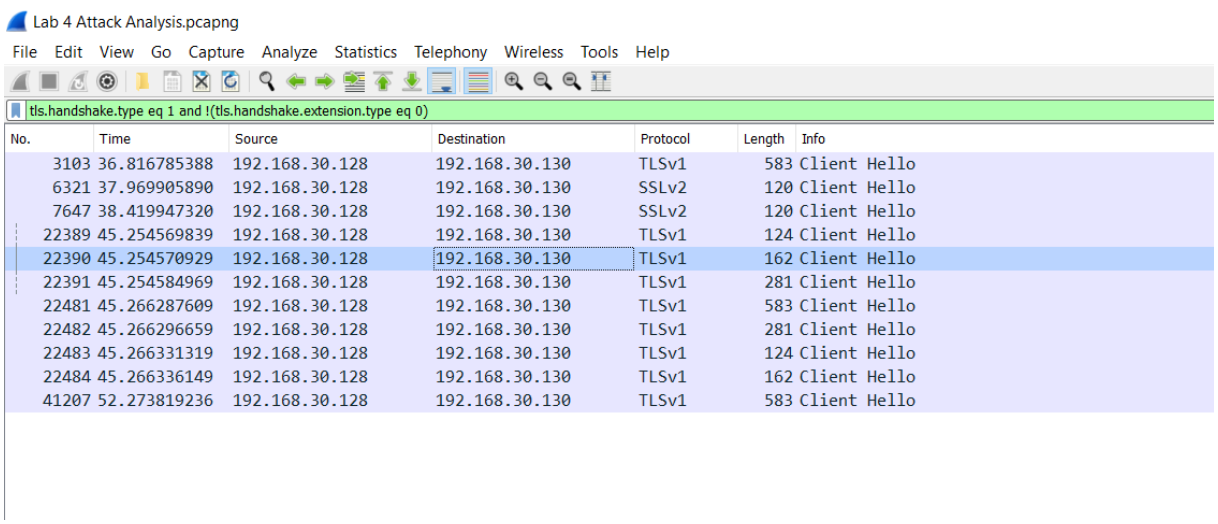

Figure A6. TLS handshake filtering results showing encrypted-session negotiation activity between internal endpoints.

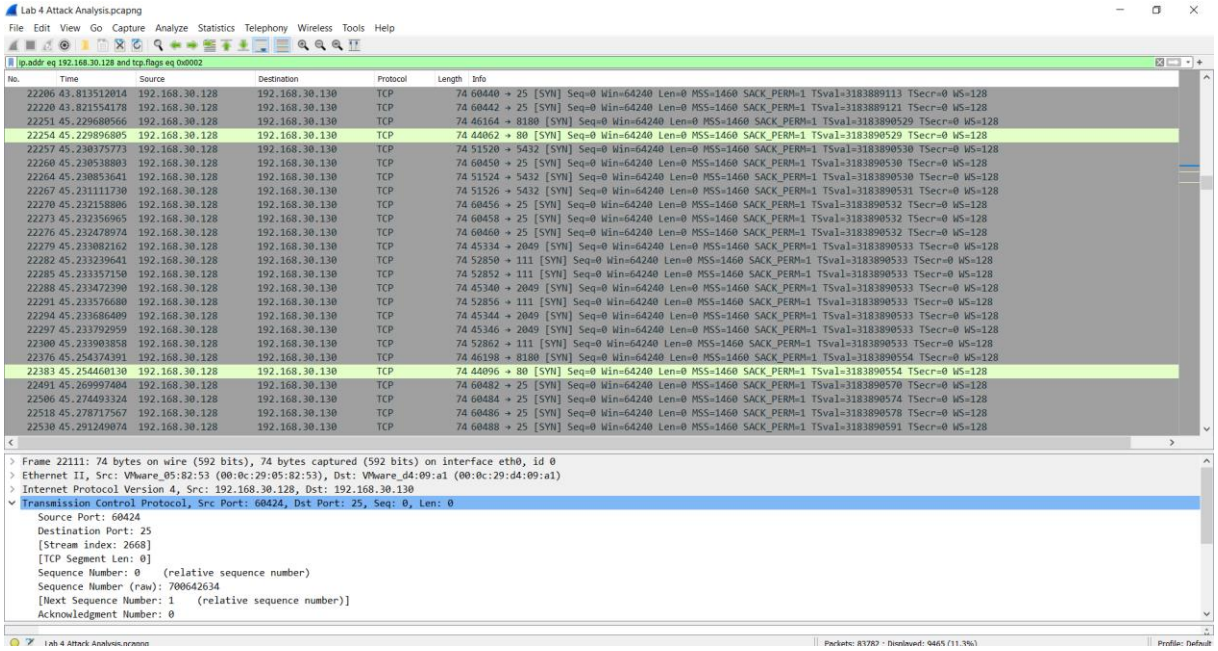

Figure A7. TCP filtering results showing connection attempts from 192.168.30.128 to multiple ports on 192.168.30.130, supporting session and conversation analysis.

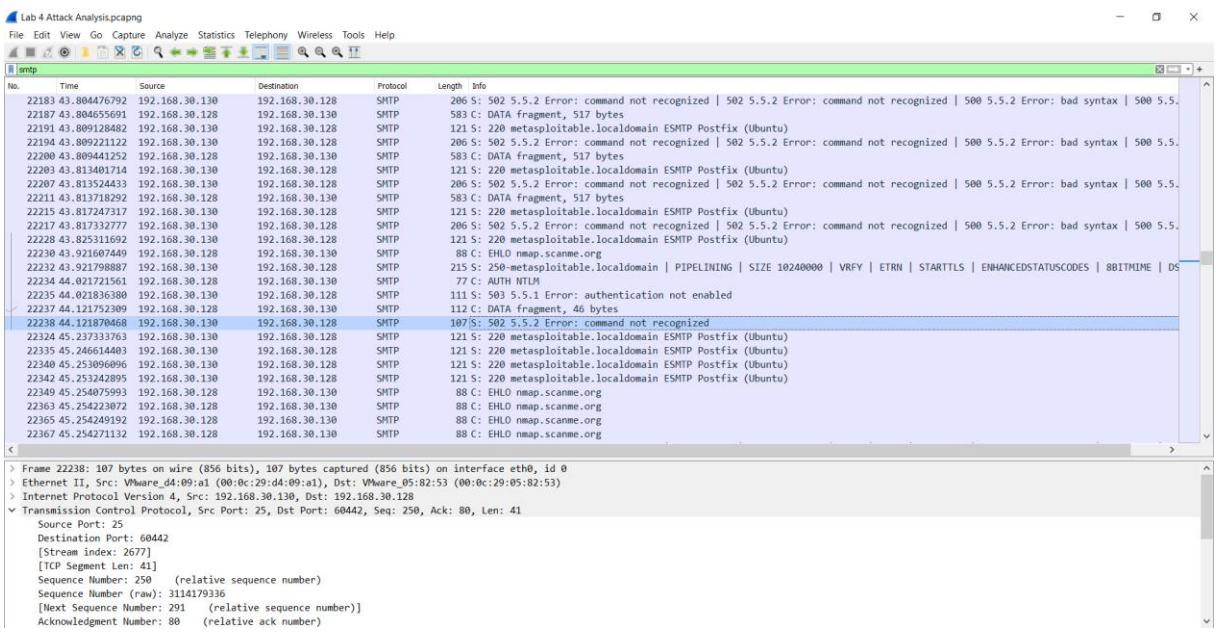

Figure A8. SMTP filtering results showing repeated mail-related communication attempts, EHLO activity, and server-side error responses.

Lab 4 Attack Analysis.pcapng

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tls.handshake.type eq 1 and !(tls.handshake.extension.type eq 0)

Wireshark · Endpoints · Lab 4 Attack Analysis.pcapng

Ethernet · 15 | IPv4 · 10 | IPv6 · 5 | TCP · 6954 | UDP · 2963

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	AS Number	AS Organization
192.168.30.1	195	37k	195	37k	0	0	—	—	—	—
192.168.30.2	6,212	540k	3,106	270k	3,106	270k	—	—	—	—
192.168.30.128	77,295	10M	41,969	7112k	35,326	3389k	—	—	—	—
192.168.30.130	83,501	11M	38,430	3659k	45,071	7382k	—	—	—	—
192.168.30.254	4	1350	2	684	2	666	—	—	—	—
192.168.30.255	38	3657	0	0	38	3657	—	—	—	—
224.0.0.22	5	300	0	0	5	300	—	—	—	—
224.0.0.251	28	2352	0	0	28	2352	—	—	—	—
224.0.0.252	8	542	0	0	8	542	—	—	—	—
239.255.255.250	118	31k	0	0	118	31k	—	—	—	—

Figure A9. IPv4 endpoint summary identifying the most active source and destination addresses in the packet capture.

Lab 4 Attack Analysis.pcapng

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arp and eth.dst eq ff:ff:ff:ff:ff:ff

No.	Time	Source	Destination	Protocol	Length	Info
35	25.172362783	VMware_fe:ef:5a	Broadcast	ARP	60	Who has 192.168.30.128? Tell 192.168.30.2
2156	25.254482806	VMware_d4:09:a1	Broadcast	ARP	60	Who has 192.168.30.2? Tell 192.168.30.130
83460	437.878681185	VMware_d4:09:a1	Broadcast	ARP	60	Who has 192.168.30.254? Tell 192.168.30.130
83530	559.255839972	VMware_d4:09:a1	Broadcast	ARP	60	Who has 192.168.30.2? Tell 192.168.30.130

```
> Frame 2156: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0
> Ethernet II, Src: VMware_d4:09:a1 (00:0c:29:d4:09:a1), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Address Resolution Protocol (request)
```

Figure A10. ARP request filtering results showing broadcast address-resolution traffic on the local segment.